Microbiology: Viral Infections of the Respiratory System

NEW YORK INSTITUTE OF TECHNOLOGY

Joseph Holloway, Ph.D., M.Sc.

Department of Biomedical Sciences

Email: jhollo03@nyit.edu



Do.
Make.
Innovate.
Reinvent the Future.

Learning Objectives

NEW YORK INSTITUTE OF TECHNOLOGY

By the end of this lecture, the student will be able to

1. Identify the properties of the most common viral pathogens of the respiratory tract, including picornaviruses, adenoviruses, influenza virus, respiratory syncytial virus and coronaviruses.
2. Describe the pathogenesis of each virus.
3. Identify the etiology, clinical presentation, and treatment for conditions associated with viral respiratory pathogens.
4. Describe the epidemiology involved in the spread and acquisition of respiratory viral diseases.
5. Outline the laboratory techniques for the identification and diagnosis of viral pathogens.

Viral Infections of the Respiratory System

NEW YORK INSTITUTE OF TECHNOLOGY

- Picornaviruses: Rhinoviruses, Enteroviruses
- Adenoviruses
- Orthomyxoviruses: Influenza virus
- Paramyxoviruses: Parainfluenza, Respiratory Syncytial Virus (RSV)
- Coronaviruses: SARS-CoV-2 (COVID-19)

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology of Respiratory Viruses

- Modes of transmission
  - Direct transmission
    - Large droplets (droplet transmission)
    - Person-to-person
  - Indirect transmission
    - Vehicles: fomites, food, water
    - Airborne: Small droplets (droplet nuclei/aerosols)
    - Vectors: living organisms

Picornaviruses

NEW YORK INSTITUTE OF TECHNOLOGY

- Non-enveloped, icosahedral nucleocapsid, (+) ssRNA
- Four major polypeptides VP1-VP4
  - Surface capsid proteins VP1 and VP3: Ab-binding sites

Enteroviruses

- Summer and fall
- <5% of common colds in adults annually

- Replicate optimally at 37C
- Stable at pH 3.0/bile for 1-3 hours
  - Polioviruses, Coxsackieviruses, Hepatitis A, Echovirus

Rhinoviruses

- Early fall and late spring (prevalence is LOWEST in summer)
- 30-50% of common colds in adults annually

- Replicate optimally at 33C
- Acid labile
- >100 human antigenic types

NEW YORK INSTITUTE OF TECHNOLOGY

Coxsackievirus A: Hand-Foot-Mouth Disease (HFMD)

Epidemiology

- Stool and respiratory secretions
- Fecal-oral, droplets, fomites, vesicular fluid
- Viral shed oropharynx < 4 weeks, stool 6 weeks
- Summer and early fall (Aug-Oct), age < 7 years
- Incubation period 2-7 days

Pathogenesis

- Replicate in submucosal lymphoid tissue of lower intestine and pharynx
- Recovered from blood and throat culture in early stages, vesicular fluid

NEW YORK INSTITUTE OF TECHNOLOGY

Coxsackievirus A: Hand-Foot-Mouth Disease (HFMD)

Clinical Features

- Non-pruritic rash of hands and feet, painful oral ulcers, fever (<101F)
- Self-limited, resolves 7-10 days



Diagnosis and Treatment

- Clinical symptoms
- PCR
- TX: supportive
- PVN: hand hygiene



Rhinovirus

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- Self-inoculation: virus deposited on hands or surface then transported to nose/eyes
- Viral shedding up to 3 weeks
- Children are major reservoir
- Late fall and early spring (LOW in summer)
- Incubation period 1-4 days

Pathogenesis

- Nasopharynx initial site of infection
- Replication limited to surface epithelium of nasal mucosa (ICAM-1)
- Pro-inflammatory signaling occurs

Rhinovirus

NEW YORK INSTITUTE OF TECHNOLOGY

Clinical Features

- Sneezing, rhinorrhea, nasal obstruction, sore throat, dry cough, headache (in children, fever)
- Resolves 5-7 days in adults, 10 days in children

Diagnosis and Treatment

- Clinical symptoms
- PCR
- TX: supportive

Adenoviruses

NEW YORK INSTITUTE OF TECHNOLOGY

- Non-enveloped, icosahedral nucleocapsid, linear, dsDNA
- >60 serotypes (7 associated with respiratory disease)
- Environmentally stable; <5% of common colds in adults

Patient population	Clinical syndromes
Infants	Pharyngitis, coryza, otitis media, pneumonia, diarrhea
Children	Upper respiratory disease, pneumonia, pharyngoconjunctival fever, diarrhea, mesenteric adenitis, hemorrhagic cystitis
Adults	ARDS, epidemic keratoconjunctivitis

Adenovirus

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- Direct contact, respiratory droplets, fecal-oral, contaminated fomites
- Viral shedding 1-7 days for adults
- Viral shedding 3-6 weeks for children
- Circulate year-round
- Incubation 2-14 days

Infection

- Hexon: group-reactive antigenic determinant
- Protein fiber and penton base: binds cell surface receptor CAR for internalization
- Early regulatory proteins prevent apoptosis/suppress IFNa and MHC-I

Adenovirus

NEW YORK INSTITUTE OF TECHNOLOGY

Clinical Features

- Fever, malaise, headache, abdominal pain, pharyngitis, coryza, conjunctivitis, tonsillitis, pneumonia
- Epidemic keratoconjunctivitis, pharyngoconjunctival fever
- Resolves in 5-7 days



Diagnosis and Treatment

- PCR
- Immunofluorescence for antibody detection
- TX: supportive



Orthomyxoviruses

NEW YORK INSTITUTE OF TECHNOLOGY

- Enveloped, helical nucleocapsid, segmented, (-) ssRNA

- Types A-C
  - Type A: antigenically highly variable
    - Infectious to humans, birds, and swine
    - Antigenic drift AND antigenic shift
  - Type B: may exhibit antigenic changes
  - Type C: antigenically stable; NO neuraminidase gene

- 5-15% of common colds in adults annually

Influenza A

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- Direct contact, aerosols, contaminated surfaces
- Viral shedding 1 day before and for 5-10 days after symptom onset
- Seasonal epidemic (Jan-April); global pandemic potential
- Incubation period 1-4 days

Infection

- Hemagglutinin: spike protein for viral entry
- Neuraminidase: cleaves sialic acid for viral exit; lowers mucus viscosity to promote lower respiratory infection

NEW YORK INSTITUTE OF TECHNOLOGY

Influenza A

Clinical Features

- Abrupt onset fever, dry cough, myalgia (malaise, sore throat, nausea, headache, nasal congestion)
- Children: higher fever, gastrointestinal issues

Diagnosis and Treatment

- Clinical symptoms
- PCR
- TX: neuraminidase inhibitors; supportive



Indicates croup; Much more frequent with parainfluenza

NEW YORK INSTITUTE OF TECHNOLOGY

Paramyxoviruses

- Enveloped, helical nucleocapsid, pleomorphic, linear, (-) ssRNA
- Hemagglutinin-neuraminidase and fusion glycoproteins: antigenic targets for neutralizing antibody
- 10% of common colds in adults annually (Parainfluenza virus (PIV) + RSV)

Paramyxoviridae

- Respirovirus (PIV 1 and 3)
- Rubulavirus (Mumps, PIV 2 and 4)
- Morbillivirus (Measles)
- Henipavirus (Hendra, Nipah)

Pneumoviridae

- Pneumovirus (RSV)
- Metapneumovirus (Human metapneumovirus)

Parainfluenza (PIV) in Children

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- Direct contact, aerosols, fomites
- Viral shedding 1 week after onset
- PIV-1, 2, seasonal in fall/winter
- PIV-3 spring
- PIV-4 causes mild URI
- Infects children < 5 years old
- Incubation period 2-6 days

Clinical Features

- Rhinitis, pharyngitis, fever, laryngotracheobronchitis (croup)
- Lower: bronchitis, pneumonia, bronchiolitis

Diagnosis and Treatment

- Clinical symptoms
- PCR
- Tx: supportive

RSV in Children

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- Large droplet direct contact, aerosols, fomites
- Viral shedding 1-3 weeks
- Reinfection is common, presents as the common cold
- Seasonal outbreaks (Oct/Nov-April/May)
- Incubation period 4-6 days

Pathogenesis

- Replicates in nasopharynx → small bronchiolar airways
- Restricted to respiratory epithelium (shed apically into airway lumen)
- Airway obstruction due to inflammation, mucus, and cell debris

RSV in Children

NEW YORK INSTITUTE OF TECHNOLOGY

Clinical Features

- URI (older kids): cough, coryza, rhinorrhea, conjunctivitis, otitis media
- LRI (infants): pneumonia, bronchiolitis

Diagnosis and Treatment

- Rapid antigen tests
- PCR
- TX: ribavirin, antiviral monoclonal Abs; supportive

NEW YORK INSTITUTE OF TECHNOLOGY

Coronaviruses

- Enveloped, helical nucleocapsid, linear, (+) ssRNA
- Largest genome among RNA viruses
- 7 serotypes that can infect humans (see below)
- 10-15% of common colds in adults annually

Alphacoronaviruses

- HCoV-229E
- HCoV-NL63

Betacoronaviruses

- HCoV-OC43
- HCoV-HKU1
- SARS-CoV-1 (Severe Acute Respiratory Syndrome)
- MERS-CoV (Middle East Respiratory Syndrome)
- SARS-CoV-2 (COVID-19)

SARS-CoV-2 (COVID-19)

NEW YORK INSTITUTE OF TECHNOLOGY

Epidemiology

- 20-40% asymptomatic infections
- Close contact with respiratory droplets, contaminated surfaces, fomites
- Viral shedding for 7-10 days
- Year-round; becoming seasonal
- Incubation period 3-5 days (up to 14 days)

Pathogenesis

- Spike protein: binds ACE2 receptor of host cells
- Cellular protease cleaves spike leading to viral entry
- Receptors found on many tissues, notably the respiratory epithelium

SARS-CoV-2 (COVID-19)

NEW YORK INSTITUTE OF TECHNOLOGY

Clinical Features

- Cough, sore throat, congestion, runny nose, and shortness of breath
- Fever, chills, malaise, and fatigue

- Complications: ARDS, Long covid

Diagnosis and Treatment

- PCR
- Antigen-detection at home
- Serology for antibody detection
- TX: supportive

Summary

NEW YORK INSTITUTE OF TECHNOLOGY

- Viral respiratory infections are common in people of all ages
- Viral respiratory infections are usually self-limited but can produce severe clinical manifestations in at-risk populations
- The spread of such viruses is generally through direct and indirect transmission routes
- Diagnosing viral respiratory infections is usually done clinically, by antigen testing, or PCR
- Prevention is through hand hygiene, contact precautions, and vaccination (if applicable)

References

NEW YORK INSTITUTE OF TECHNOLOGY

- Chin-Hong P. et al. 2026. Levinson's Review of Medical Microbiology & Immunology, 19th ed. McGraw-Hill Education.
- Murray, PR et. al. 2026. Medical Microbiology, 10th ed. Elsevier.
- Tao, LE et. al. 2026. First Aid for the USMLE Step 1; McGraw Hill Publication.
- UptoDate:
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/respiratory-syncytial-virus-infection-in-infants-and-children-clinical-features-and-diagnosis
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/covid-19-epidemiology-virology-and-prevention
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/parainfluenza-viruses-in-children
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/seasonal-influenza-in-adults-clinical-manifestations-and-diagnosis
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/pathogenesis-epidemiology-and-clinical-manifestations-of-adenovirus-infections
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/hand-foot-and-mouth-disease-and-herpangina
- https://nyit.idm.oclc.org/login?url=https://www-uptodate-com.nyit.idm.oclc.org/contents/epidemiology-clinical-manifestations-and-pathogenesis-of-rhinovirus-infections

THANK YOU!

NEW YORK INSTITUTE OF TECHNOLOGY

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R